GWCD Documentation - ETL Pipeline

Vermont AHS GIS and Data Viz | Groundwater Chemistry Database (GWCD)

ETL Pipeline Walkthrough

Source: `GroundwaterChemistryDatabasePROD.py` (runs weekly). Line numbers below refer to the sanitized copy in scripts/.

1. Load raw data

Truncate and reload `GroundwaterChemistry_GWCDrawdata` from `groundwater.ChemistryResults` (straight column copy, no filtering).

2. Weekly exceedance email alert

Filters the last 7 days of records against a fixed set of per-analyte concentration thresholds (metals, VOCs, BNAs, PCBs, herbicides, PFOS/PFAS) and emails a CSV + HTML summary to an internal distribution list. This is a notification tool, separate from the public map/dashboard data below.

3. Data cleaning

- Zip code fixes (leading zeros, dash artifacts).
- `WaterSource` normalization (e.g. "dug well" → "Dug Well"; unrecognized values backfilled from `SamplingLocationDescription` where possible, otherwise set to "Unknown").
- Town name cleanup via the alias table (typos/out-of-state variants mapped to a canonical municipality where known).
- `AnalyteName` → `AnalyteAlias` standardization (e.g. "arsenic total" → "Arsenic").

4. Geocoding

- Addresses are concatenated (`Address1, City, VT, Zip`) and geocoded against the state's E911 composite locator.
- Records that fail to geocode an address are retried using **just the town name**, geocoded to a **town centroid**.
- Records that fail both are dropped from the geocoded/mapped product (they still exist in the raw table from step 1).

- All geocoded points (address-level + centroid fallback) are spatially joined to town boundaries to assign `TOWNNAME`.

See 05-geocoding.md for details and what this means for state- vs. town-level analysis.

5. Scope filter

Before any exceedance math, records are restricted to the private drinking water scope: valid/non-error results, private well/spring water sources only (excludes public/municipal/surface water), non-institutional clients (schools, daycares, VDH internal proficiency tests excluded), and non-post-treatment samples. Full logic in 03-filtering-criteria.md.

6. Exceedance calculation

Per record, per applicable threshold type (MCL / SMCL / VHA / general Threshold), flag whether the result is above or below. Non-detects are represented as half the reporting limit. Full logic in 04-exceedance-calculations.md.

7. Deduplication

For a given (Address, Analyte, Town) combination, only one record is kept — the highest result (ties broken by most recent report date). This prevents repeat testing at one location from skewing town-level statistics.

8. Town/analyte aggregation

Grouped by (`AnalyteAlias` , `TOWNNAME`), compute: sample count, count/percent above each threshold type, and mean/median/min/max/95th percentile/standard deviation of the result value.

9. Write outputs

- Full set (deduplicated, with stats) → `GroundwaterChemistry_GWCDPoints`.
- Public set → `GroundwaterChemistry_GWCDPointTableExternal` : same data, with town/analyte groups under 20 samples suppressed, and extra human-readable popup fields generated (`MapMean` , `MapPopupHeading` , etc.) for 6 specific analytes (Arsenic, Fluoride, Gross Alpha, Nitrate, Radon, Uranium) — see 06-application.md.

10. Housekeeping

Rebuilds table indexes, updates a "last data refresh" timestamp in a SharePoint list (via Microsoft Graph API), and publishes a pivoted (long-to-wide) CSV export of all results to ArcGIS Online for internal use.